

ONLINE MASTERS IN **DATA SCIENCE**

DSC258: NATURAL LANGUAGE PROCESSING

NER AND DICTIONARIES

UC San Diego

COMPUTER SCIENCE & ENGINEERING
HALICIOĞLU DATA SCIENCE INSTITUTE

Classic Named Entity Recognition (NER)

Some questions to think about:

- Can massive raw texts help? → Neural language model
- **Can dictionaries help?**
- Are human annotations always correct?
- Is Tokenizer always good?

Input

- Unlabeled Raw Texts
- An Entity Dictionary
 - entity type, canonical name, [synonyms_1, synonyms_2, ..., synonyms_k]

Output

- A NER model to recognize the entities of the entity types appeared in the given dictionary.
- Note that the entities to be recognized can be unseen entities.

String-match / rule-based distant supervision generation

AutoEntity, SwellShark, ClusType, ...

- Leave the entity span detection to experts
- POS Tag Rule-based (e.g., regular expressions)

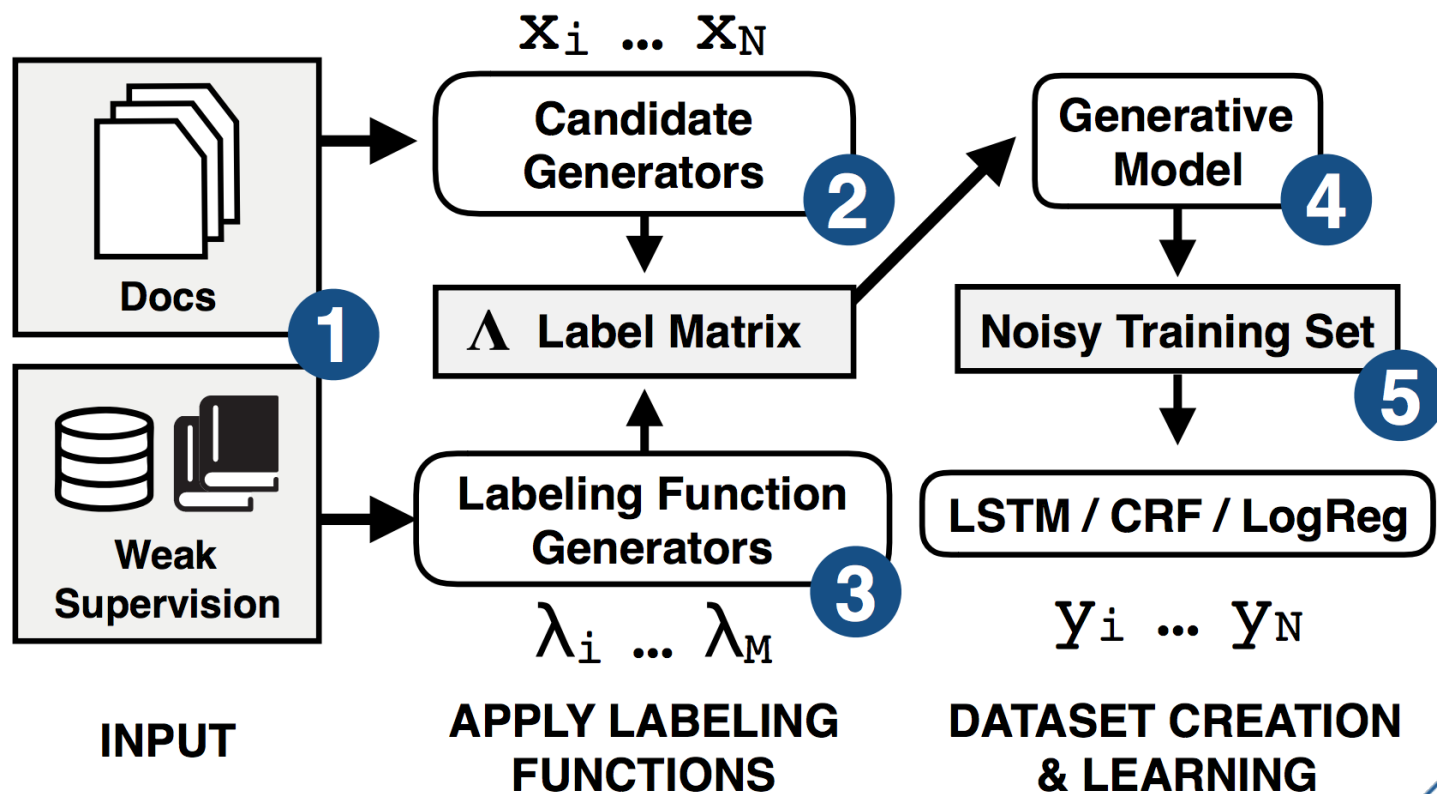
AutoNER

- A novel “Tie-or-Break” labeling scheme + tailored neural model

Data Programming for Typing

Entity Span Detection: Regular expressions based on part-of-speech (POS) tags

- **Requires expert efforts**
- Candidate Generators



A core dictionary

- Leads to high-precision but low-recall matches

A “full” dictionary

- Leads to high-recall but low-precision matches
- Introduce out-of-dictionary high-quality phrases as new entities
 - Their types are “unknown”
 - It could be any BIOES + any type

AUTONER: FUZZY-LSTM-CRF BASELINE

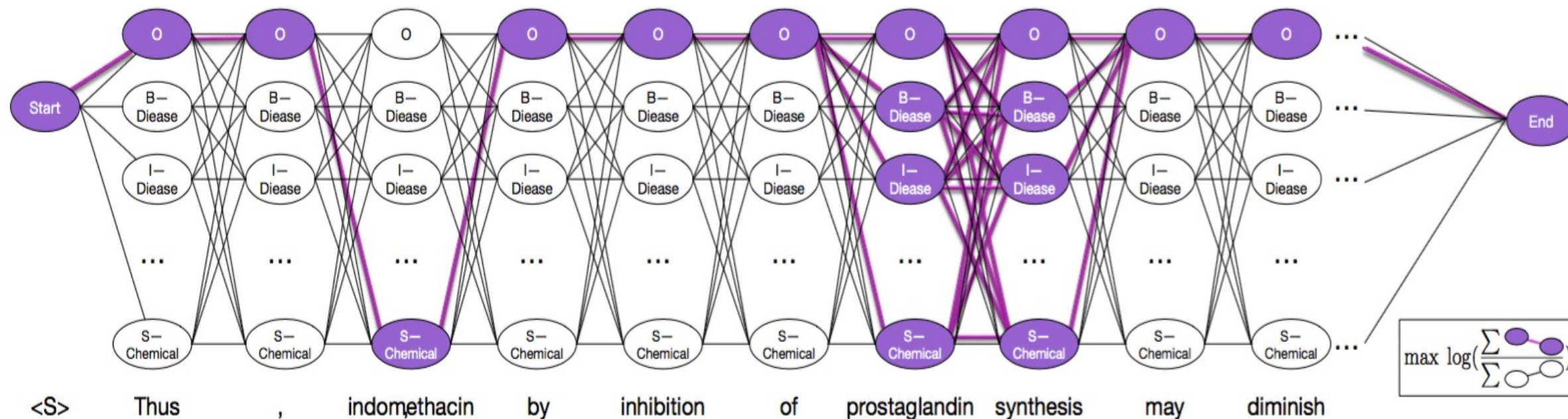


Figure 1: The illustration of the Fuzzy CRF layer with modified IOBES tagging scheme. The named entity types are {Chemical, Disease}. "indomethacin" is a matched Chemical entity and "prostaglandin synthesis" is an unknown-typed high-quality phrase. Paths from Start to End marked as purple form all possible label sequences given the distant supervision.

Instead of labeling each token, we choose to tag the connection between two adjacent tokens.

For every two adjacent tokens, the connection between them is labeled as

- (1) **Tie**, when the two tokens are matched to the same entity
- (2) **Unknown**, if at least one of the tokens belongs to an unknown-typed high-quality phrase;
- (3) **Break**, otherwise.

AUTONER: TAILORED NEURAL MODEL

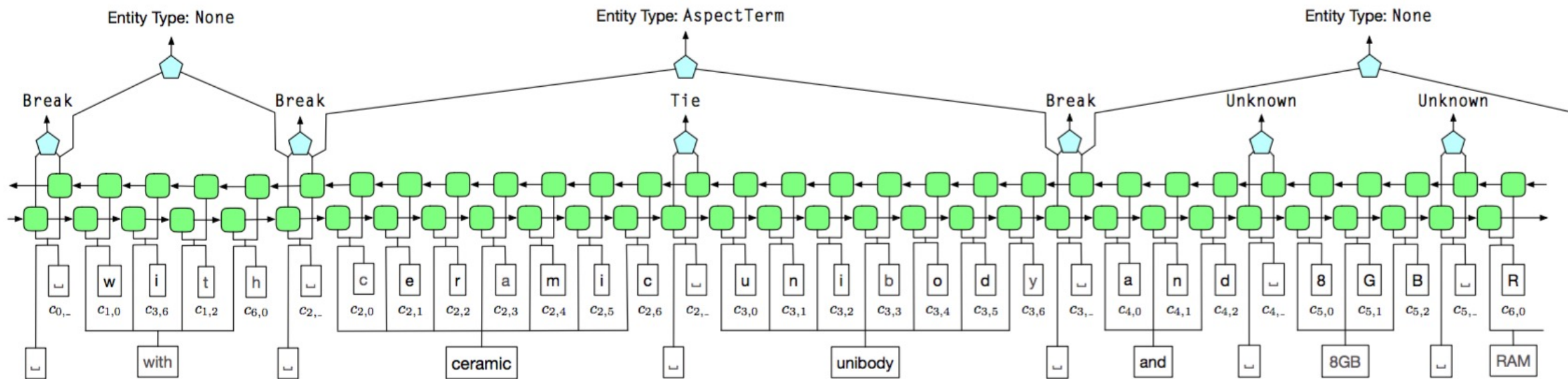


Figure 2: The illustration of AutoNER with Tie or Break tagging scheme. The named entity type is {AspectTerm}. “ceramic unibody” is a matched AspectTerm entity and “8GB RAM” is an unknown-typed high-quality phrase. Unknown labels will be skipped during the model training.

Table 2: [Biomedical Domain] NER Performance Comparison. The supervised benchmarks on the BC5CDR and NCBI-Disease datasets are LM-LSTM-CRF and LSTM-CRF respectively (Wang et al., 2018). SwellShark has no annotated data, but for entity span extraction, it requires pre-trained POS taggers and extra human efforts of designing POS tag-based regular expressions and/or hand-tuning for special cases.

Method	Human Effort other than Dictionary	BC5CDR			NCBI-Disease		
		Pre	Rec	F1	Pre	Rec	F1
Supervised Benchmark					85.49	85.80	
SwellShark					80.1	80.8	
Dictionary					69.7	67.1	
Fuzzy-LSTM					56.15	69.32	
AutoNER					67.71	73.28	
					71.98	75.52	

AutoNER $\approx$ 300 expert annotated articles!
(in biomedical domain)

Classic Named Entity Recognition (NER)

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- Can massive raw texts help? → Neural language model
- **Can dictionaries help? → Distant supervised setting**
- Are human annotations always correct?
- Is Tokenizer always good?